

Interest rate derivatives

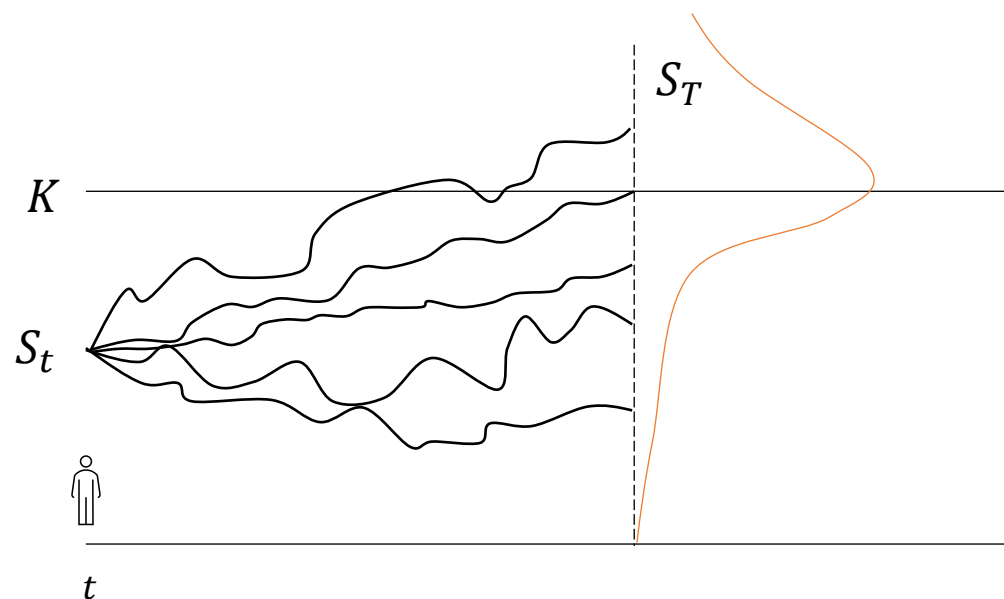
2. Non-Linear Derivatives

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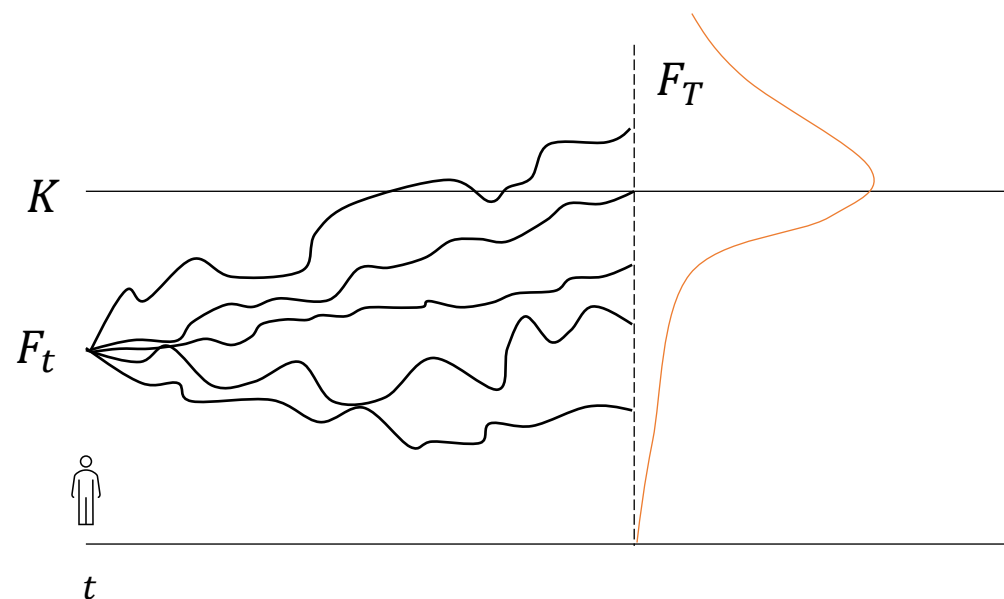
The Familiar Black Scholes for Stock Options



- ▶ In BSM World, the interest rate is constant ($= r$), and the stock price follows lognormal distribution.
- ▶ The payoff is given as $(S_T - K)^+$. The Price of a call option at time (t) is given by

$$c(t) = S_t \mathcal{N}(d_1) - K e^{-r(T-t)} \mathcal{N}(d_2) , d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + (r + 0.5\sigma^2)(T-t)}{\sigma\sqrt{T-t}} , d_2 = \frac{\ln\left(\frac{S_t}{K}\right) + (r - 0.5\sigma^2)(T-t)}{\sigma\sqrt{T-t}}$$

The Familiar Black Scholes for Stock Options



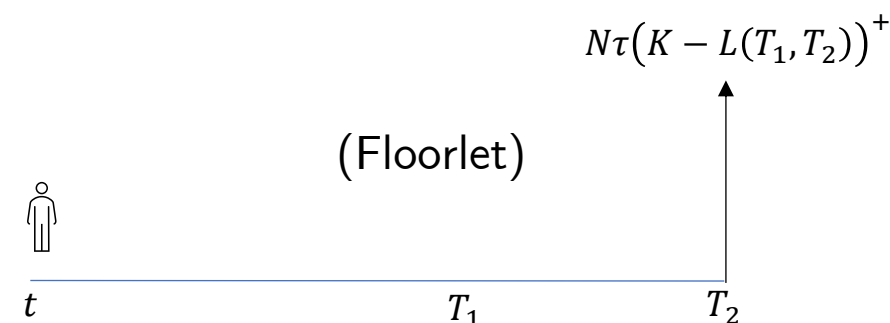
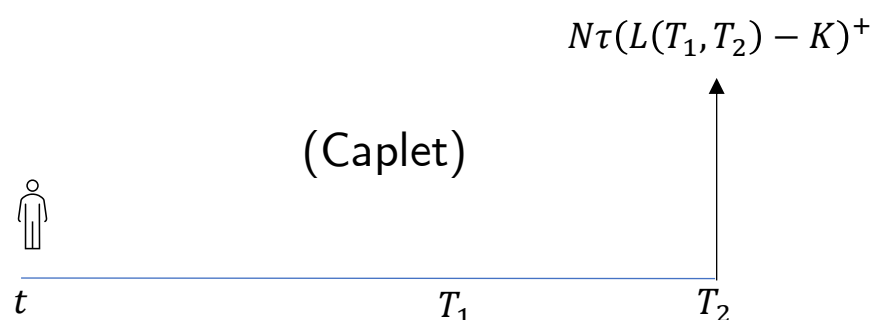
► It can also be expressed in terms of the Forward Price of the stock: $F(t, T) = S_t e^{r(T-t)}$

$$c(t) = e^{-r(T-t)} (F_t \mathcal{N}(d_1) - K \mathcal{N}(d_2)) , d_1 = \frac{\ln\left(\frac{F_t}{K}\right) + (0.5 * \sigma^2)(T-t)}{\sigma \sqrt{T-t}} , d_2 = \frac{\ln\left(\frac{F_t}{K}\right) - (0.5 * \sigma^2)(T-t)}{\sigma \sqrt{T-t}}$$

Recognize that $\sigma^2(T-t)$ is the terminal variance of the stock price, let's denote it as $v^2 = \sigma^2(T-t)$
We can re-write the formula as

$$c(t) = e^{-r(T-t)} (F_t \mathcal{N}(d_1) - K \mathcal{N}(d_2)) , d_1 = \frac{\ln\left(\frac{F_t}{K}\right) + (0.5 * v^2)}{v} , d_2 = \frac{\ln\left(\frac{F_t}{K}\right) - (0.5 * v^2)}{v}$$

Interest Rate Caplets / Floorlets



- ▶ An interest rate Caplet has a payoff $:= N\tau \max((L(T_1, T_2) - K), 0)$
- ▶ The value of the Caplet is given by Black76 model where the forward rate $F(t, T_1, T_2)$ is the underlier which follows lognormal distribution

$$\text{Caplet}(t) = N\tau P(t, T_2) \left(F_t \mathcal{N}(d_1) - K \mathcal{N}(d_2) \right) , \quad d_1 = \frac{\ln\left(\frac{F_t}{K}\right) + (0.5 * v^2)}{v} , \quad d_2 = \frac{\ln\left(\frac{F_t}{K}\right) - (0.5 * v^2)}{v}$$

Similarly, the price of a Floorlet is given as

$$\text{Floorlet}(t) = N\tau P(t, T_2) \left(-F_t \mathcal{N}(-d_1) + K \mathcal{N}(-d_2) \right) , \quad d_1 = \frac{\ln\left(\frac{F_t}{K}\right) + (0.5 * v^2)}{v} , \quad d_2 = \frac{\ln\left(\frac{F_t}{K}\right) - (0.5 * v^2)}{v}$$

Interest Rate Caplets / Floorlets

- We can develop a neat hierarchical function that gives the values of Caplet or Floorlet

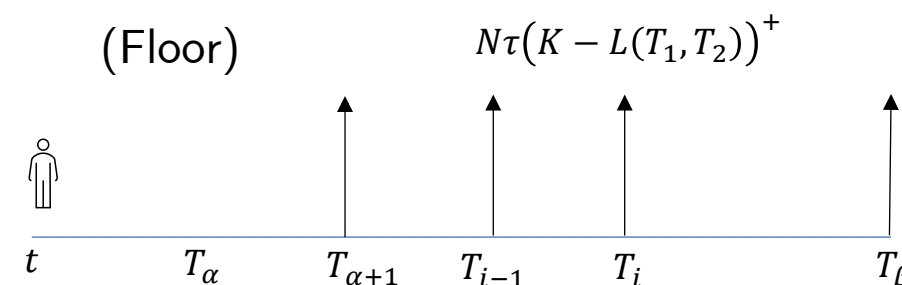
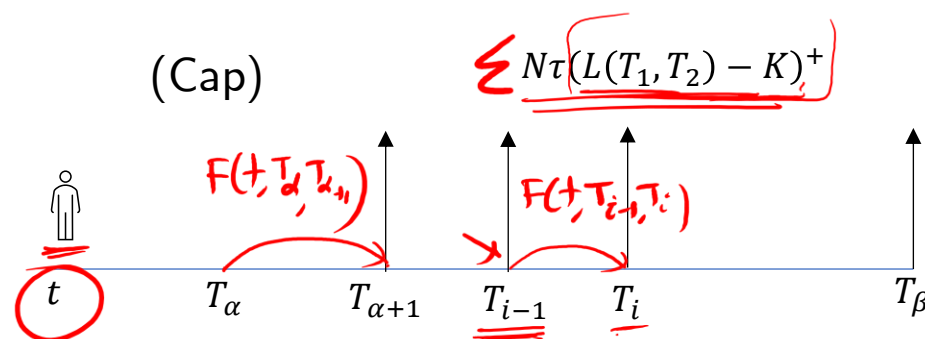
$$\text{Bl}(K, F, v, \omega) = F\omega\mathcal{N}(\omega d_1(K, F, v)) - K\omega\mathcal{N}(\omega d_2(K, F, v)), \quad \omega = 1(\text{Caplet}), -1(\text{Floorlet})$$

$$d_1(K, F, v) = \frac{\ln\left(\frac{F}{K}\right) + \left(\frac{1}{2}v^2\right)}{v}, \quad d_2(K, F, v) = \frac{\ln\left(\frac{F}{K}\right) - \left(\frac{1}{2}v^2\right)}{v}$$

$$v = \sigma\sqrt{T_1 - t}$$

$$V(t, T_1, T_2, F, \tau, K, \sigma) = N\tau P(t, T_2)\text{Bl}(K, F, v, \omega)$$

Interest Rate Caps / Floors



- Caps and Floors are essentially a series of Caplets and Floorlets. Thankfully, these products can be simply decomposed into individual Caplets/Floorlets, and the value of the series is the sum of these individual options. The formula for Cap/Floor is given as

$$V(t, T, F, \tau, K, \sigma) = N\tau \sum_{i=\alpha+1}^{\beta} P(t, T_i) \text{Bl}(K, \underline{F(t, T_{i-1}, T_i)}, \underline{v_i}, \omega)$$

$\sigma \sqrt{T_{i-1} - t}$

$$\underline{v_i} = \sigma \sqrt{T_{i-1} - t}$$

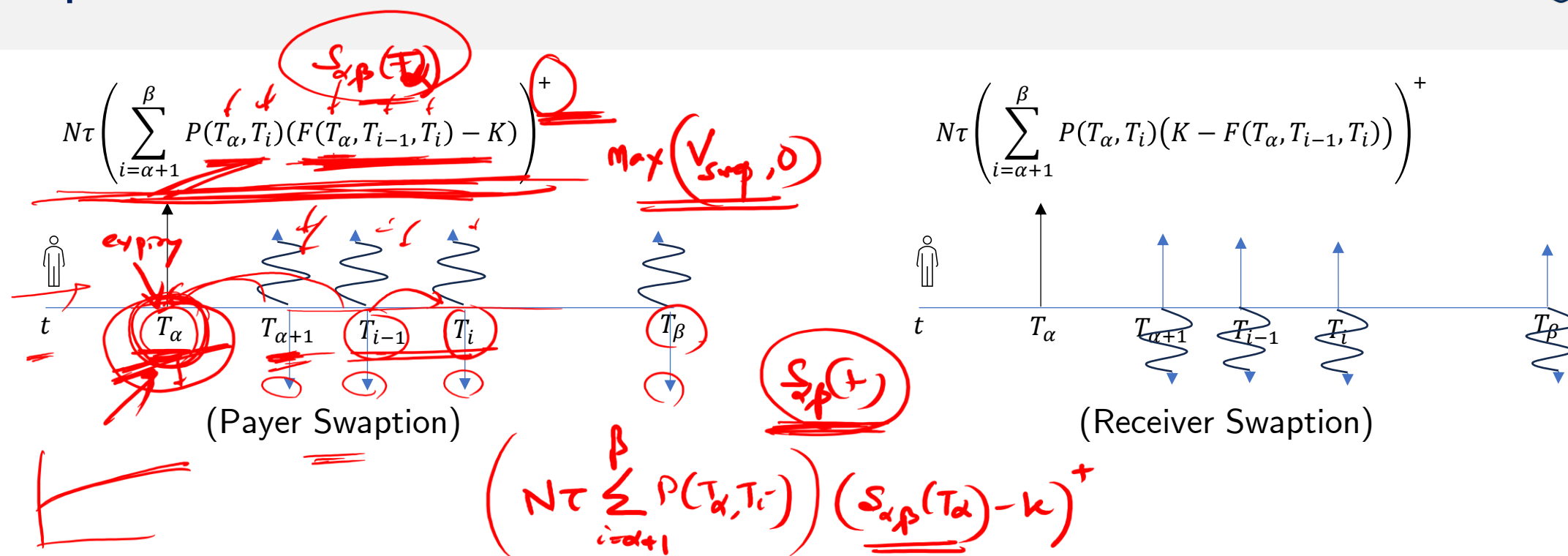
- Yes, this formula suffers from the same drawbacks as Black Scholes such as constant volatility assumption. Just as in case of equity we have volatility surface i.e., volatility depends on (time to maturity and moneyness), in case of Caplets/Floorlets we have volatility cubes i.e., volatility depends on (tenor, expiry, moneyness)

- Yes, put call parity holds. Caplet – Floorlet = FRA

$$S_T - k$$

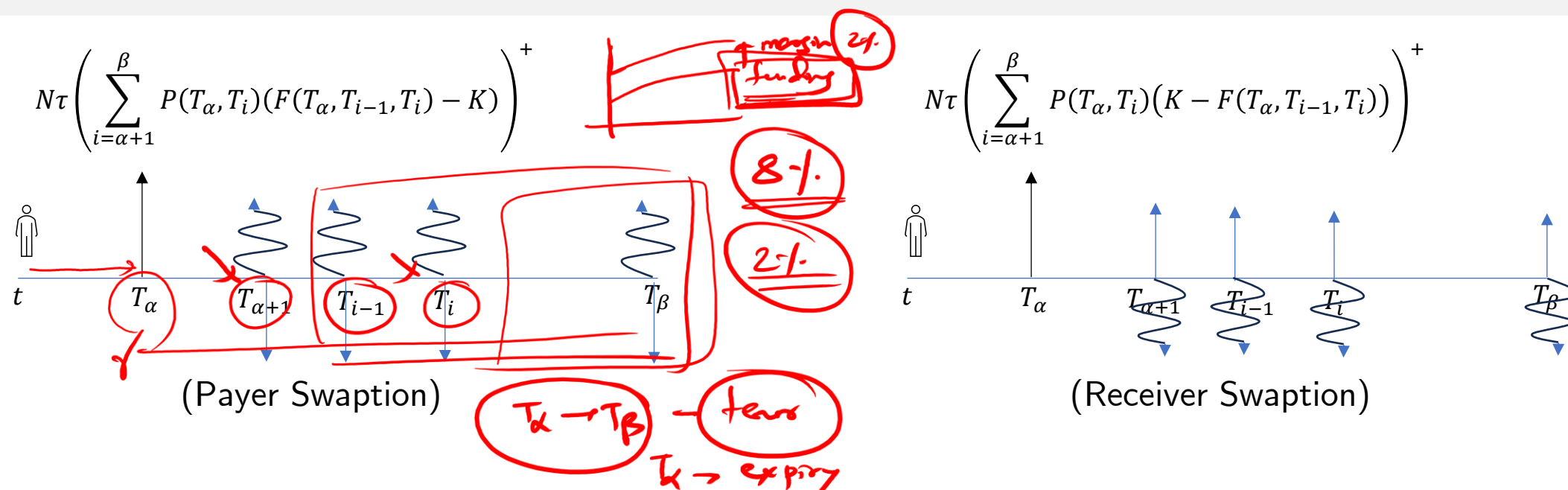
$\underline{T_1, T_2}$
 $\textcircled{D}$ $\frac{F(t; T_{i-1}, T_i) - K}{T}$
 $(L(T_1, T_2) - K)^+ - (K - L(T_1, T_2))^+ = (L - K)$
 - Floorlet = FRA
 $\frac{\textcircled{S_T}}{T} - (K - \underline{S_T})^+ = (S_T - K)^+ = (L - K)$
 $\textcircled{T}$
 $= (S_T - K), S_T \geq K$
 $= -(K - S_T), S_T \leq K$

Swaptions



- ▶ A swaption gives the holder the right to enter a swap at the expiry date (T_α). Of course, the option will be exercised when the swap's valuation is +ve at expiry (T_α)
- ▶ If the option is exercised and the swaption is physically settled, then the holder owns the underlying swap after option expiry ; if it's cash settled then only the value at expiration is settled.

Swaptions



► The valuation of swaption is done using Black76 formula where the underlier is the swap rate $s_{\alpha, \beta}(t)$

$$\text{Swaption}^{Black}(t, T, \tau, N, K, \sigma, \omega) = N\tau \text{Bl}\left(K, s_{\alpha, \beta}(t), \sigma \sqrt{T_{\alpha} - t}, \omega\right) \sum_{i=\alpha+1}^{\beta} P(t, T_i)$$

Handwritten notes include: $\sigma = \sigma \sqrt{T_{\alpha} - t}$, $(sN(\theta_1) - KN(\theta_2))$, and "Annuity".

$\omega = +1$ (Payer Swaption), -1 (Receiver Swaption)

Swaptions

- ▶ Between Cap and Payer Swaption, which will have higher price ? Ans – Cap
- ▶ Is there a put call parity version for swaption ? Yes.

Payer Swaption – Receiver Swaption = Forward Payer Swap

$$\begin{aligned}
 & \begin{array}{c} (f-k) \\ \uparrow \\ x \end{array} - \begin{array}{c} (k-f) \\ \uparrow \\ z \end{array} \\
 & \begin{array}{c} x \\ \uparrow \\ 1 \end{array} \quad \begin{array}{c} y \\ \uparrow \\ 1 \end{array} \quad \begin{array}{c} z \\ \uparrow \\ 1 \end{array} \\
 & \hline
 & x^+ + y^+ + z^+ \\
 & \hline
 & \text{Cap} \\
 & \hline
 & 100 - 100 \\
 & \hline
 & 100 + 0 \\
 & \hline
 & \rightarrow \\
 & \begin{array}{c} 0 \\ \downarrow \\ 0 \end{array} \\
 & \hline
 & \begin{array}{c} (x+y+z)^+ \\ \hline \text{Swaption} \end{array} \\
 & \hline
 & \begin{array}{c} \max(V_{\text{swap}}^f, 0) \\ \hline \max(-V_{\text{swap}}^f, 0) = V_{\text{swap}} \end{array}
 \end{aligned}$$

$V_{\text{swap}} = N \tau \frac{(f-k)}{(k-f)}$